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Inventor(s)	Komatsu; Atsushi et al.

Processing method

Abstract

A cutting method that measures heights of an outer periphery of a wafer, calculates height distribution data, and then calculates change rates of the heights of the outer periphery of the wafer. Based on results of a comparison between the height change rates and a threshold, the cutting method determines whether or not foreign matter exists on a back surface of the wafer (i.e., whether or not foreign matter exists on a holding surface of a chuck table). The existence of foreign matter on the holding surface can hence be detected appropriately. If foreign matter exists on the holding surface, a worker is notified, and the cutting operation is cancelled. Therefore, the cutting method can appropriately give notification to the worker that foreign matter is stuck on the holding surface, and can also appropriately avoid performing cutting processing with foreign matter stuck on the holding surface.

Inventors:	**Komatsu; Atsushi (Tokyo, JP), Makino; Koichi (Tokyo, JP)**
Applicant:	**DISCO CORPORATION** (Tokyo, JP)
Family ID:	**91963879**
Assignee:	**DISCO CORPORATION (Tokyo, JP)**
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Primary Examiner: Sanchez; Omar Flores

Attorney, Agent or Firm: GREER BURNS & CRAIN, LTD.

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a cutting apparatus and a cutting method.

Description of the Related Art

(2) A silicon wafer with a plurality of devices such as integrated circuits (ICs) or large scale integration (LSI) formed on a front surface thereof is ground at a back surface thereof to be formed to have a predetermined thickness, and is then divided into individual device chips by a cutting apparatus. The divided device chips are used in electronic equipment such as mobile phones and personal computers.

(3) When dividing a workpiece such as the above-described silicon wafer into individual device chips, a cutting apparatus that includes a chuck table, which has a holding surface that holds the workpiece by suction, and a cutting blade is used in general (see, for example, JP 2000-087282A).

The cutting blade is, for example, a hub blade having a hub and a blade electrodeposited on the hub by a Ni plating process.
(4) On the cutting apparatus, cutting processing for the workpiece is performed by the cutting blade with the workpiece held under suction on the chuck table.

SUMMARY OF THE INVENTION

(5) With a cutting apparatus as described above, the workpiece may be damaged when held on the chuck table if foreign matter is stuck on the holding surface of the chuck table. It is therefore a current practice that before holding a workpiece on a chuck table, a holding surface is imaged by an imaging unit to inspect whether or not foreign matter exists on the holding surface (see, for example, JP 2000-216227A).
(6) However, foreign matter may be small in size, and may hence be difficult to detect even by the imaging unit. Even if the workpiece is not damaged in this situation, the workpiece may be bulged at a part thereof by the foreign matter, and if subjected to cutting processing as it is, its processing quality may be reduced.
(7) The present invention therefore has as objects thereof the provision of a cutting apparatus and a cutting method, which can efficiently detect foreign matter on a holding surface.
(8) In accordance with a first aspect of the present invention, there is provided a cutting apparatus for cutting a chamfered outer peripheral portion of a workpiece. The cutting apparatus includes a chuck table having a holding surface that holds the workpiece, a processing unit that processes the workpiece held on the chuck table, a height measurement unit that measures heights of the workpiece held on the chuck table, and a controller that controls at least the chuck table, the processing unit, and the height measurement unit of the cutting apparatus. The controller includes a height distribution calculation section that calculates height distribution data of an outer periphery of the workpiece by measuring the heights of the workpiece at locations inside by a predetermined distance from an outer peripheral edge portion thereof with the height measurement unit while rotating the chuck table at least a full turn, a change rate calculation section that calculates height change rates as change rates of the heights of the outer periphery in the workpiece, based on the height distribution data calculated by the height distribution calculation section, a determination section that determines, based on results of comparison between the height change rates and a threshold, whether or not foreign matter exists on a back surface of the workpiece, and an error notification section that gives notification of an error if the foreign matter is determined to exist on the back surface of the workpiece.
(9) In accordance with a second aspect of the present invention, there is provided a cutting method for cutting a chamfered outer peripheral portion of a workpiece. The cutting method includes a holding step of holding the workpiece by a chuck table having a holding surface, a height distribution calculation step of, after the holding step, calculating height distribution data of an outer periphery of the workpiece by measuring heights of the workpiece at locations inside by a predetermined distance from an outer peripheral edge portion thereof while rotating the chuck table at least a full turn, a change rate calculation step of calculating height change rates as change rates of the heights of the outer periphery in the workpiece, based on the height distribution data calculated in the height distribution calculation step, a determination step of determining, based on results of comparison between the height change rates and a threshold, whether or not foreign matter exists on a back surface of the workpiece, and a processing step of notifying an error if the foreign matter is determined to exist in the determination step or performing cutting of the chamfered outer peripheral portion if no foreign matter is determined to exist in the determination step.
(10) In the present invention, the heights of the outer periphery of the workpiece are measured to calculate the height distribution data and then to calculate the change rates of the heights of the outer periphery of the workpiece. Based on the results of the comparison between the height change rates and the threshold, a determination is made as to whether or not foreign matter exists

on the back surface of the workpiece, in other words, whether or not foreign matter exists on the holding surface of the chuck table. In the present invention, the sticking of foreign matter on the holding surface can hence be detected appropriately.
(11) If foreign matter exists on the back surface of the workpiece, a notification is made accordingly in this invention. It is therefore possible to appropriately communicate, for example, to a worker that the foreign matter is stuck on the holding surface.
(12) The above and other objects, features and advantages of the present invention and the manner of realizing them will become more apparent, and the invention itself will best be understood from a study of the following description and appended claims with reference to the attached drawings showing some preferred embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view depicting a configuration of a cutting apparatus according to an embodiment of a first aspect of the present invention;
(2) FIG. **2** is a cross-sectional view depicting configurations of a chuck table and one of two height measurement units in the cutting apparatus of FIG. **1**;
(3) FIG. **3** is a perspective view depicting a height distribution calculation step in a cutting method according to an embodiment of the second aspect of the present invention;
(4) FIG. **4**A is a cross-sectional view depicting foreign matter stuck on a part of a front surface of an outer periphery of a wafer, and FIG. **4**B is a cross-sectional view depicting foreign matter stuck on a part of a back surface of the outer periphery of the wafer;
(5) FIG. **5** is a graph illustrating two examples of height distribution data in the cases of FIGS. **4**A and **4**B that the foreign matter is stuck on the front surface and back surface of the wafers, respectively;
(6) FIG. **6** is a graph illustrating an example of height distribution data in a case that no foreign matter is stuck on a wafer;
(7) FIG. **7** is a graph illustrating an example of height distribution data in the case that the foreign matter is stuck on the front surface of the wafer; and
(8) FIG. **8** is a graph illustrating an example of height distribution data in the case that the foreign matter is stuck on the back surface of the wafer.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(9) With reference to FIGS. **1** and **2**, a description will first be made about a cutting apparatus **1** according to an embodiment of the first aspect of the present invention. FIG. **1** is a perspective view depicting the configuration of the cutting apparatus **1**, and FIG. **2** is a cross-sectional view depicting the configurations of a chuck table **20** and one of two height measurement units **30** in the cutting apparatus **1** of FIG. **1**. A wafer **100** depicted in FIG. **1** is an example of a workpiece, and has a substantially circular shape as seen in plan. On a front surface **101** of the wafer **100**, a plurality of devices **103** is formed. The wafer **100** has a back surface **102**, which is a surface to be ground by grinding stones or the like.
(10) As depicted in FIG. **2**, the wafer **100** has been chamfered at an outer peripheral edge thereof, and a chamfered outer peripheral portion **105** having an arc-shaped cross-section is formed at the outer peripheral edge. The cutting apparatus **1** depicted in FIG. **1** is useful in cutting (performing edge-trimming of) the chamfered outer peripheral portion **105** of the wafer **100**.
(11) The cutting apparatus **1** is a generally-called dual dicing saw. The cutting apparatus **1** includes a bed **10**, a gantry column **14** disposed upright on the bed **10**, and a controller **7** that controls individual elements of the cutting apparatus **1**.
(12) On the bed **10**, an X-axis direction moving mechanism **11** is arranged. The X-axis direction

moving mechanism **11** moves the chuck table **20** in a cutting feed direction (X-axis direction). The X-axis direction moving mechanism **11** includes a pair of guide rails **111** extending in the X-axis direction, an X-axis table **113** mounted on the guide rails **111**, a ball screw **110** extending in parallel to the guide rails **111**, and a motor **112** that rotates the ball screw **110**.

(13) The paired guide rails **111** are arranged in parallel to the X-axis direction on an upper surface of the bed **10**. On the paired guide rails **111**, the X-axis table **113** is slidably arranged along the guide rails **111**. On the X-axis table **113**, a holding unit **2** is mounted with the chuck table **20** included therein.

(14) The ball screw **110** is in threaded engagement with a nut portion (not depicted) disposed on the X-axis table **113**. The motor **112** is connected to one end portion of the ball screw **110** to rotationally drive the ball screw **110**. By the rotational drive of the ball screw **110**, the X-axis table **113** and holding unit **2** are moved in the X-axis direction along the guide rails **111**.

(15) The holding unit **2** includes the chuck table **20** and a θ table **25**. The chuck table **20** has a holding surface **22** that holds the wafer **100**. The θ table **25** is a rotary shaft that rotates with the chuck table **20** supported thereon.

(16) As depicted in FIG. **2**, the chuck table **20** has a substantially disk-shaped frame body **23**, and in a recessed portion disposed in the frame body **23**, also has a holding member **21** made from a porous material such as porous ceramic. The holding member **21** has an upper surface, which serves as the holding surface **22** that holds the wafer **100** by suction. In the chuck table **20**, the holding member **21** is brought into communication with a suction source (not depicted), whereby the wafer **100** can be held under suction on the holding surface **22**.

(17) As depicted in FIG. **1**, the chuck table **20** is supported on the θ table **25** arranged on a side of a bottom surface of the chuck table **20**. The θ table **25** is disposed rotatably in an X-Y plane on an upper surface of the X-axis table **113**. Therefore, the θ table **25** supports the chuck table **20**, and further can rotationally drive the chuck table **20** in the X-Y plane.

(18) On a rear side (−X direction side) on the bed **10**, the gantry column **14** is disposed upright so as to extend astride the X-axis direction moving mechanism **11**.

(19) On a front surface (a surface on a +X direction side) of the gantry column **14**, a cutting unit moving mechanism **13** is arranged to move the first cutting unit **18** and a second cutting unit **19**. The cutting unit moving mechanism **13** performs index feeding of the first cutting unit **18** and second cutting unit **19** in a Y-axis direction, and also their cutting-in feeding in a Z-axis direction.

(20) The cutting unit moving mechanism **13** includes a first Z-axis direction moving mechanism **16**, a second Z-axis direction moving mechanism **17**, and a Y-axis direction moving mechanism **12**. The first Z-axis direction moving mechanism **16** moves the first cutting unit **18**, which is on a +Y direction side, in the Z-axis direction. The second Z-axis direction moving mechanism **17** moves the second cutting unit **19**, which is on a −Y direction side, in the Z-axis direction. The Y-axis direction moving mechanism **12** moves the first cutting unit **18** and second cutting unit **19** in the Y-axis direction.

(21) The Y-axis direction moving mechanism **12** is arranged on the front surface of the gantry column **14**. The Y-axis direction moving mechanism **12** reciprocally moves the first Z-axis direction moving mechanism **16**, including the first cutting unit **18**, and the second Z-axis direction moving mechanism **17**, including the second cutting unit **19**, along the Y-axis direction.

(22) The Y-axis direction moving mechanism **12** includes a pair of guide rails **121** extending in the Y-axis direction, a first Y-axis table **123** and a second Y-axis table **125** attached to the guide rails **121**, a first ball screw **120** and second ball screw **122** extending in parallel to the guide rails **121**, a first motor **124** that rotates the first ball screw **120**, and a second motor (not depicted) that rotates the second ball screw **122**.

(23) In parallel to the Y-axis direction, the paired guide rails **121** are arranged on the front surface of the gantry column **14**. On the paired guide rails **121**, the first Y-axis table **123** and a second Y-axis table **125** are slidably arranged along these guide rails **121**. On the first Y-axis table **123**, the

first Z-axis direction moving mechanism **16** and first cutting unit **18** are attached. On the second Y-axis table **125**, the second Z-axis direction moving mechanism **17**, and second cutting unit **19** are attached.
(24) The first ball screw **120** is in threaded engagement with a nut portion (not depicted) disposed on the first Y-axis table **123**. The first motor **124** is connected to one end portion of the first ball screw **120**, and rotationally drives the first ball screw **120**. By the rotational drive of the first ball screw **120**, the first Y-axis table **123**, first Z-axis direction moving mechanism **16**, and first cutting unit **18** are moved in the Y-axis direction along the guide rails **121**.
(25) Similarly, the second ball screw **122** is in threaded engagement with a nut portion (not depicted) disposed on the second Y-axis table **125**, and is rotationally driven by the second motor connected to one end portion of the second ball screw **122**. As a consequence, the second Y-axis table **125**, second Z-axis direction moving mechanism **17**, and second cutting unit **19** are moved in the Y-axis direction along the guide rails **121**.
(26) The first Z-axis direction moving mechanism **16** reciprocally moves the first cutting unit **18** in the Z-axis direction. The first Z-axis direction moving mechanism **16** includes a pair of guide rails **161** extending in the Z-axis direction, a support member **163** arranged on the guide rails **161**, a ball screw **160** extending in parallel to the guide rails **161**, and a motor **162** that rotates the ball screw **160**.
(27) On the first Y-axis table **123**, the paired guide rails **161** are arranged in parallel to the Z-axis direction. On the paired guide rails **161**, the support member **163** is slidably arranged along these guide rails **161**. On a lower end portion of the support member **163**, the first cutting unit **18** is attached.
(28) The ball screw **160** is in threaded engagement with a nut portion (not depicted) disposed on the support member **163**. The motor **162** is connected to one end portion of the ball screw **160**, and rotationally drives the ball screw **160**. By the rotational drive of the ball screw **160**, the support member **163** and first cutting unit **18** are moved in the Z-axis direction along the guide rails **161**.
(29) The second Z-axis direction moving mechanism **17** also reciprocally moves the second cutting unit **19** in the Z-axis direction. The second Z-axis direction moving mechanism **17** has a similar configuration as the first Z-axis moving mechanism **16**, and its description is omitted accordingly.
(30) The first cutting unit **18** and second cutting unit **19** are examples of processing units that process the wafer **100** held on the chuck table **20**. The first cutting unit **18** serves to cut the wafer **100** held on the chuck table **20** and has a cutting blade **181** and an imaging unit **182**. The imaging unit **182** images the wafer **100** held on the holding surface **22** of the chuck table **20**, and detects the chamfered outer peripheral portion **105** as a location where the wafer **100** is to be cut. The cutting blade **181** is used to cut the chamfered outer peripheral portion **105** of the wafer **100**.
(31) Similarly to the first cutting unit **18**, the second cutting unit **19** also serves to cut the wafer **100** held on the chuck table **20**. The second cutting unit **19** has a similar configuration as the first cutting unit **18**, and its description is omitted accordingly.
(32) On the lower end portions of the support members **163** in the first Z-axis moving mechanism **16** and second Z-axis direction moving mechanism **17**, the height measurement units **30** are attached via the support arms **40**, respectively.
(33) The height measurement units **30** are arranged such that they are located adjacent the first cutting unit **18** and second cutting unit **19**. Along with the first cutting unit **18** and second cutting unit **19**, the height measurement units **30** can be moved in the Y-axis direction and Z-axis direction by the cutting unit moving mechanism **13** that includes the Y-axis direction moving mechanism **12**, first Z-axis direction moving mechanism **16**, and second Z-axis direction moving mechanism **17**.
(34) Each height measurement unit **30** is configured to measure heights of the wafer **100** held on the chuck table **20**. Described specifically, as depicted in FIG. **2**, each height measurement unit **30** has, for example, an irradiation unit **31**, a position adjustment unit **32**, an oscillation unit **36**, and a light-receiving unit **37**. The irradiation unit **31** is arranged in a casing **35**, and applies a laser beam

L. The position adjustment unit **32** adjusts the position of the irradiation unit **31** along the Y-axis direction. The oscillation unit **36** controls the irradiation of the laser beam L from the irradiation unit **31**. The light-receiving unit **37** receives reflection light of the laser beam L.
(35) The position adjustment unit **32** includes a ball screw **33** extending in the Y-axis direction, and a motor **34** that rotates the ball screw **33**. The ball screw **33** is in threaded engagement with a nut portion (not depicted) arranged on the irradiation unit **31**. Through rotational drive of the ball screw **33** by the motor **34**, the irradiation unit **31** is moved in the Y-axis direction.
(36) Each height measurement unit **30** having the configuration described above applies the laser beam L from the irradiation unit **31** to a vicinity of the chamfered outer peripheral portion **105** of the wafer **100** held on the holding surface **22** of the chuck table **20**, and also to a frame body surface **24** as an upper surface of the frame body **23**. Based on reflection light of the laser beam L, the height measurement unit **30** can measure the height of the frame body surface **24** and the height of the vicinity of the chamfered outer peripheral portion **105** in the wafer **100**, in other words, the height of the wafer **100** at a location inside by a predetermined distance from an outer peripheral edge portion (an outer peripheral edge of the chamfered outer peripheral portion **105**) in the wafer **100**.
(37) As depicted in FIG. **1**, the cutting apparatus **1** also includes a touch screen **8** attached to an undepicted casing. On the touch screen **8**, a variety of kinds of information on the cutting apparatus **1** are displayed. The touch screen **8** is also used to set “various kinds of other information. As appreciated from the foregoing, the touch screen **8** functions as a display device for displaying information, and also functions as an input device for inputting information.
(38) In a vicinity of the touch screen **8**, a speaker **9** is also included to transmit information to a worker by voice.
(39) The cutting apparatus **1** is also provided with the controller **7**. The controller **7** includes a central processing unit (CPU) that performs processing according to a control program, and a recording medium such as a memory. The controller **7** performs a variety of processing and controls the individual elements of the cutting apparatus **1**.
(40) The controller **7** also includes a height distribution calculation section **71**, a change rate calculation section **72**, a determination section **7**, and an error notification section **74**, and uses these sections to perform cutting on the wafer **100** while detecting foreign matter on surfaces of the wafer **100**.
(41) A description will hereinafter be made about a cutting method according to an embodiment of the second aspect of the present invention, which is controlled by the controller **7**. This cutting method is useful in cutting the chamfered outer peripheral portion **105** of the wafer **100**. As will hereinafter be described in detail, the cutting method of this embodiment includes a holding step, a height distribution calculation step, a change rate calculation step, a determination step, and a processing step.
(42) (1) Holding Step
(43) In this step, the wafer **100** is held on the chuck table **20** having the holding surface **22**. Described specifically, the worker or an undepicted transfer device places, in this step, the wafer **100** on the holding surface **22** of the chuck table **20** in the holding unit **2**. The controller **7** then controls the undepicted suction source, so that a suction force is transmitted to the holding surface **22**. As a consequence, the wafer **100** is held under suction on the holding surface **22**.
(44) (2) Height Distribution Calculation Step
(45) With reference to FIG. **3**, the height distribution calculation step will next be described. FIG. **3** is a perspective view depicting the height distribution calculation step. The controller **7** controls the X-axis direction moving mechanism **11**, cutting unit moving mechanism **13**, and θ table **25**, and as depicted in FIG. **3**, sets the position of the height measurement unit **30** such that the laser beam L from the height measurement unit **30** is applied to the chamfered outer peripheral portion **105** of the wafer **100** and a vicinity thereof.

(46) While controlling the θ table **25** and rotating the chuck table **20** at least a full turn, the height distribution calculation section **71** of the controller **7** subsequently controls the height measurement unit **30**, so that heights of the wafer **100** at locations inside by the predetermined distance from the outer peripheral edge portion of the wafer **100** are measured as heights of an outer periphery of the wafer **100** along the outer periphery of the wafer **100**. Based on the heights measured as described above, the height distribution calculation section **71** calculates height distribution data of the outer periphery of the wafer **100**. These pieces of height distribution data are those which indicate “the distribution of heights of the wafer **100** along the outer periphery of the wafer **100**.”
(47) Now, a description will be made about two examples of a relation between foreign matter stuck on the wafer **100** and the height distribution data. FIG. **4**A is a cross-sectional view depicting foreign matter F stuck on a part of the front surface **101** in the outer periphery of the wafer **100**. In the case of FIG. **4**A, heights Ta measured along the outer periphery of the wafer **100** are high at only the part where the foreign matter F is stuck. FIG. **5** is a graph illustrating two examples of height distribution data in the cases of FIGS. **4**A and **4**B that the foreign matter F is stuck on the front surface **101** of and back surface **102** of the wafer **100**, respectively. In the case of FIG. **4**A, the height distribution data calculated by the height distribution calculation section **71** hence also increases locally at only a measuring point, which corresponds to the foreign matter F, and its vicinity as illustrated by the graph (a) in FIG. **5**.
(48) Meanwhile, FIG. **4**B is a cross-sectional view depicting foreign matter F stuck on a part of the back surface **102** in the outer periphery of the wafer **100**, in other words, on the holding surface **22** of the chuck table **20**. In the case of FIG. **4**B, heights Tb measured along the outer periphery of the wafer **100** gradually increase toward the part where the foreign matter F is stuck. In the case of FIG. **4**B, the height distribution data calculated by the height distribution calculation section **71** also gently increases toward a measuring point corresponding to the foreign matter F as illustrated by the graph (b) in FIG. **5**.
(49) (3) Change Rate Calculation Step
(50) Based on the height distribution data calculated by the height distribution calculation section **71** in the height distribution calculation step, the change rate calculation section **72** of the controller **7** next calculates height change rates, that is, the change rates of the heights along the outer periphery in the wafer **100**.
(51) Here, the change rate calculation section **72** acquires, based on the height distribution data, the heights at the measuring points along the outer periphery of the wafer **100**. The change rate calculation section **72** then finds each height change rate, for example, using the following equation (1):
Height change rate=((change amount of height between two measuring points)/(distance between two measuring points))×100%   (1)
(52) where the change amount of height between two measuring points means, for example, “the difference in measured height between the two measuring points,” and the distance between two measuring points is, for example, “the distance between the two points along the outer periphery of the wafer **100**.”
(53) Assume, for example, that four measuring points A to D are set side by side in this order on the outer periphery of the wafer **100**. The locations of the measuring points A to D on the outer periphery of the wafer **100** are assumed to be r.sub.A, r.sub.B, r.sub.C, and r.sub.D, respectively, and the heights measured at the measuring points A to D are assumed to be H.sub.A, H.sub.B, H.sub.C, and H.sub.D, respectively.
(54) In this case, using the height change amount (H.sub.B−H.sub.A) between A and B, the height change amount (H.sub.C−H.sub.B) between B to C, the height change amount (H.sub.D−H.sub.C) between C to D, the distance (r.sub.B−r.sub.A) between A and B, the distance (r.sub.C−r.sub.B) between B to C, and the distance (r.sub.D−r.sub.C) between C to D, the height change rate between A and B, the height change rate between B to C, and the height change rate between C to D are

calculated, respectively, as follows:

Height change rate between A and B=((H.sub.B−H.sub.A)/(r.sub.B−r.sub.A))×100%

Height change rate between B to C=((H.sub.C−H.sub.B)/(r.sub.C−r.sub.B))×100%

Height change rate between C to D=((H.sub.D−H.sub.C)/(r.sub.D−r.sub.C))×100%

(4) Determination Step

(55) Based on the results of comparison between the height change rates calculated by the change rate calculation section **72** and predetermined thresholds, the determination section **73** of the controller **7** next determines whether or not foreign matter exists on the back surface **102** of the wafer **100**.

(56) In this embodiment, two threshold values, one being a relatively small first threshold value and the other a relatively large second threshold value, are set for determination by the determination section **73**. Using these two thresholds, the determination section **73** determines that no foreign matter is stuck on the wafer **100**, foreign matter is stuck on the front surface **101** of the wafer **100**, or foreign matter is stuck on the back surface **102** of the wafer **100**.

(57) Examples of determination by the determination section **73** will hereinafter be described using the example of the above-mentioned four measuring points A to D. FIG. **6** is a graph illustrating an example of height distribution data in a case that no foreign matter is stuck on the wafer **100**. As illustrated in FIG. **6**, the heights H.sub.A, H.sub.B, H.sub.C, and H.sub.D measured at the four measuring points A to D have substantially the same value. Therefore, the maximum value among the height change rate between A and B, the height change rate between B to C, and the height change rate between C to D is very small, that is, is smaller than the relatively small first threshold value. The determination section **73** hence determines that no foreign matter is stuck on the wafer **100** if the maximum value of height change rate as determined by the change rate calculation section **72** is smaller than the first threshold value.

(58) Now assume that foreign matter is stuck in a vicinity of the measurement point D on the front surface **101** of the wafer **100**. In this case, the height of the outer periphery of the wafer **100** locally increases near a location where foreign matter exists, as described using FIGS. **4**A and **5**. FIG. **7** is a graph illustrating examples of height distribution data in a case that foreign matter is stuck on the front surface **101** of the wafer **100**. In this case, as illustrated in FIG. **7**, the measured height of the outer periphery of the wafer **100** hence locally increases at the measuring point D and its adjacent measuring point C, but decreases at the measuring points A and B apart from the measuring point D. The maximum value among a height change rate between A and B, a height change rate between B to C, and a height change rate between C to D is therefore greater than the relatively large second threshold value. The determination section **73** thus determines that foreign matter is stuck on the front surface **101** of the wafer **100** if the maximum value among the height change rates as determined by the change rate calculation section **72** is greater than the second threshold value.

(59) Also assume that foreign matter is stuck in a vicinity of the measuring point D on the back surface **102** of the wafer **100**. In this case, the height of the outer periphery of the wafer **100** gradually increases toward a part where foreign matter is stuck, as described using FIGS. **4**B and **5**. FIG. **8** is a graph illustrating examples of height distribution data in a case that foreign matter is stuck on the back surface **102** of the wafer **100**. In this case, as illustrated in FIG. **8**, the measured heights of the outer periphery of the wafer **100** change such that they gently increase toward the measuring point D. The maximum value among a height change rate between A and B, a height change rate between B to C, and a height change rate between C to D is therefore greater than the first threshold value, but smaller than the second threshold value. The determination section **73** thus determines that foreign matter exists on the back surface **102** of the wafer **100** (in other words, foreign matter exists on the holding surface **22** of the chuck table **20**) if the maximum value among the height change rates as determined by the change rate calculation section **72** is greater than the first threshold value but smaller than the second threshold value.

(60) (5) Processing Step

(61) The error notification section **74** of the controller **7** next controls notification of an error based on the results of the determination by the determination section **73** in the determination step. If the determination section **73** determines that foreign matter exists on the back surface **102** of the wafer **100**, for example, the error notification section **74** gives notification of an error to the worker. Described specifically, using the speaker **9** depicted in FIG. **1**, the error notification section **74** gives notification of (communicates to) the worker by voice or sound (for example, an alarm sound) that foreign matter is stuck on the back surface **102** of the wafer **100**, and at the same time, also gives notification of the worker of the same by characters or an image using the touch screen **8** depicted in FIG. **1**. Then, the controller **7**, for example, cancels the cutting operation on the wafer **100**.
(62) If the determination section **73** determines that no foreign matter is stuck on the wafer **100**, on the other hand, the error notification section **74**, for example, does not perform notification of an error. In this case, the controller **7** then controls the above-mentioned individual elements in the cutting apparatus **1** to perform cutting of the chamfered outer peripheral portion **105** of the wafer **100**. Described specifically, with the chuck table **20** kept rotating by the θ table **25**, the controller **7** controls the cutting blade **181** in the first cutting unit **18** or second cutting unit **19**, so that the chamfered outer peripheral portion **105** of the wafer **100** is cut along a peripheral direction thereof.
(63) Further, if the determination section **73** determines that foreign matter is stuck on the front surface **101** of the wafer **100**, the error notification section **74** gives notification of the worker accordingly, for example, using the touch screen **8**. Similar to the case that no foreign matter is stuck on the wafer **100**, for example, the controller **7** then controls the individual elements of the cutting apparatus **1** to perform cutting of the chamfered outer peripheral portion **105** of the wafer **100**.
(64) In this embodiment, height change rates of the outer periphery of the wafer **100** are calculated by measuring heights of the outer periphery of the wafer **100** and calculating height distribution data as described above. Based on the results of comparison between the height change rates and threshold values, the determination is then made as to whether or not foreign matter exists on the back surface **102** of the wafer **100**, in other words, whether or not foreign matter exists on the holding surface **22** of the chuck table **20**. In this embodiment, it is therefore possible to appropriately detect that foreign matter is stuck on the holding surface **22**.
(65) If foreign matter exists on the back surface **102** of the wafer **100**, the worker is also notified accordingly using a notification system including the touch screen **8** and speaker **9** in this embodiment. This embodiment can therefore appropriately communicate to the worker that the foreign matter is stuck on the holding surface **22**. In this case, the controller **7** then cancels cutting operation in this embodiment. It is therefore possible to appropriately avoid performing cutting processing with foreign matter stuck on the holding surface **22**.
(66) Using the two threshold values consisting of the first threshold value and the second threshold value, the determination section **73** also determines in this embodiment that no foreign matter is stuck on the wafer **100**, foreign matter is stuck on the front surface **101** of the wafer **100**, or foreign matter is stuck on the back surface **102** of the wafer **100**. In this embodiment, it is therefore possible to determine the location (the front surface **101** or the back surface **102**) where foreign matter is stuck, in addition to the sticking of the foreign matter on the wafer **100**.
(67) It is to be noted that in this regard, the determination section **73** may be configured to determine with a single threshold value whether or not foreign matter is stuck on the back surface **102** of the wafer **100**. In this case, the determination section **73** may be configured to determine that the foreign matter exists on the back surface **102** of the wafer **100** if the maximum value among height change rates calculated by the change rate calculation section **72** is greater than the threshold value, but to determine that no foreign matter exists on the back surface **102** of the wafer **100** if the maximum value among the height change rates is equal to or smaller than the threshold value.

(68) The change rate calculation section **72** may also be configured such that in the change rate calculation step, data (a measuring point and a measured height) that are locally high are extracted from the height distribution data, and thickness change rates are then calculated using the extracted data and their preceding and trailing data. With this configuration, thickness change rates can be calculated, paying attention to a part where foreign matter is considered to be stuck on the wafer **100**. It is hence possible to efficiently calculate the thickness change rates that are useful in determining the sticking of foreign matter on the wafer **100**.
(69) The present invention is not limited to the details of the above described preferred embodiments. The scope of the invention is defined by the appended claims and all changes and modifications as fall within the equivalence of the scope of the claims are therefore to be embraced by the invention.

Claims

1. A cutting apparatus for cutting a chamfered outer peripheral portion of a workpiece, comprising: a chuck table having a holding surface that holds the workpiece; a processing unit that processes the workpiece held on the chuck table; a height measurement unit that measures heights of the workpiece held on the chuck table; and a controller that controls at least the chuck table, the processing unit, and the height measurement unit of the cutting apparatus, wherein the controller includes: a height distribution calculation section that calculates height distribution data of an outer periphery of the workpiece by measuring the heights of the workpiece at locations inside by a predetermined distance from an outer peripheral edge portion thereof with the height measurement unit while rotating the chuck table at least a full turn, a change rate calculation section that calculates height change rates as change rates of the heights of the outer periphery in the workpiece, based on the height distribution data calculated by the height distribution calculation section, a determination section that determines, based on results of comparison between the height change rates and a threshold, whether or not foreign matter exists on a back surface of the workpiece, and an error notification section that gives notification of an error if the foreign matter is determined to exist on the back surface of the workpiece.
2. A cutting method for cutting a chamfered outer peripheral portion of a workpiece, the method comprising: a holding step of holding the workpiece by a chuck table having a holding surface; a height distribution calculation step of, after the holding step, calculating height distribution data of an outer periphery of the workpiece by measuring heights of the workpiece at locations inside by a predetermined distance from an outer peripheral edge portion thereof while rotating the chuck table at least a full turn; a change rate calculation step of calculating height change rates as change rates of the heights of the outer periphery in the workpiece, based on the height distribution data calculated in the height distribution calculation step; a determination step of determining, based on results of comparison between the height change rates and a threshold, whether or not foreign matter exists on a back surface of the workpiece, and a processing step of notifying an error if the foreign matter is determined to exist in the determination step or performing cutting of the chamfered outer peripheral portion if no foreign matter is determined to exist in the determination step.